

AI FAKE INTERNSHIP THREAT SCANNER

Machine Learning-Based Threat Detection, Recruiter Verification & Risk Intelligence Platform
Review 2 to Review 3 Comprehensive Transition & Technical Audit Report

Target Evaluation: Review 3 Exhibition	Active Model: Calibrated Linear SVM	Dataset: EMSCAD (17,880 listings)
Accuracy: 98.41% (Test Set)	Fraud Precision: 90.85%	Repository: vaibhavmurty4518/fake-internship-detector

1. Executive Summary: Review 2 vs Review 3 Comparison

Between Review 2 and Review 3, the application underwent an end-to-end upgrade from a basic single-post classifier into an enterprise-grade AI cybersecurity intelligence platform. Below is the direct capability matrix:

Feature / Capability	Review 2 (Baseline)	Review 3 (Final Production)
Visual Design System	Basic light SaaS theme with flat cards and minimal hierarchy.	Cybersecurity aesthetic: Charcoal/Black palette, Frosted Glassmorphism, and live 3D WebGL Security Core.
Company & Recruiter Verification	Text-only heuristic checking without domain or webmail validation.	Integrated Multi-tier Engine: Domain registration age, typosquatted lookalike checks, corporate vs webmail audit.
Preset Test Hub	Plain text snippets without recruiter metadata; uniform styling.	6-Field Complete Presets: Unique color per scenario, auto-populating Company, Email, Recruiter, LinkedIn, URLs.
Multi-Posting Input	Not supported (single text input only).	Multi-Post Scanner Studio: Batch paste, auto-delimiter detection (---, blank lines), 3-tier risk grouping, CSV export.
UI Stability & Rendering	Vulnerable to CommonMark indentation bugs rendering raw HTML tags as text.	Zero Code-Leak Guarantee: Native <code>render_safe_html()</code> stream and isolated HTML component wrappers.
Automated Test Suite	Basic smoke tests only.	48 / 48 Automated Pytest Suite Passing across pipeline and investigation modules.

2. Machine Learning Architecture & Benchmark Performance

The classification backend leverages a **Calibrated Linear Support Vector Machine (Linear SVM)** trained on 17,880 listings from the Employment Scam Aegean Dataset (EMSCAD). Linear SVM was selected after extensive benchmarking against Logistic Regression and Random Forest due to its superior F1-score on high-dimensional sparse TF-IDF n-gram feature representations.

Model Architecture	Overall Accuracy	Fraud Precision	Fraud Recall	Fraud F1-Score
Logistic Regression (TF-IDF + Struct)	95.61%	52.68%	90.75%	0.6667
Random Forest Classifier	97.93%	100.00%	57.23%	0.7279
Linear SVM (Active Production)	98.41%	90.85%	74.57%	0.8190

True Negatives (TN): 3,390 Legitimate posts verified	False Positives (FP): 13 False alarms (0.38% rate)	False Negatives (FN): 44 Bypassed ML (caught by rules)	True Positives (TP): 129 Fraud detected accurately
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3. Recruiter Verification & Domain Investigation Pipeline

Review 3 introduces a multi-tier evidence engine that bridges textual ML predictions with live employer authentication:

- **Domain Registration & Age:** Assesses WHOIS domain age to identify newly registered ephemeral scam portals (< 6 months old).
- **Typosquatting & Lookalike Detection:** Computes Levenshtein and token distance to catch brand hijacking (e.g., microso0ft-careers.com vs microsoft.com).
- **Recruiter Email Integrity:** Flags recruiters claiming to represent Fortune 500 corporations while utilizing personal webmail domains (e.g., @gmail.com, @yahoo.com).
- **Offline Verification Cache:** Incorporates a pre-compiled JSON cache store (company_cache.json) ensuring instant, deterministic verification during offline evaluation sessions.

4. New Feature: Multi-Post Scanning Studio

A major new feature implemented for Review 3 is the **Multi-Post Scan Studio**, enabling users to analyze multiple job postings or recruitment messages concurrently without manual file creation:

- **Smart Delimiter Splitting:** Automatically detects and parses postings separated by dashes (---), double line-breaks, or header prefixes (POSTING 1:, POSTING 2:).
- **Batch Threat Summary:** Displays immediate distribution metrics across High Risk, Moderate Risk, and Low Risk categories.
- **3-Tier Grouped Risk Containers:** Presents clear, high-contrast visual groupings showing posting numbers, threat scores (/100), detected indicator tags, and raw text snippets.
- **1-Click CSV Audit Export:** Generates a complete structured CSV containing Posting Number, Threat Score, Risk Level, Fraud Probability, and Extracted Suspicious Indicators.

5. Preset Intelligence Hub (6 Complete Scenarios)

The Preset Hub has been completely overhauled with 6 real-world test cases, each fully populating all 6 metadata fields (Company, Website, Recruiter Name, Recruiter Email, Recruiter LinkedIn, Offer URL, and Description):

Scenario Title	Category & Color Theme	Threat Characteristics
Upfront Registration Fee Scam	CRITICAL THREAT (Crimson)	Demands Rs. 999 refundable registration/kit fee; contact via WhatsApp & Gmail.
Microsoft Brand Impersonation	IMPERSONATION (Cyber Blue)	Legitimate Fortune 500 company name hijacked using non-corporate webmail & fee trap.
Genuine Series-B FinTech Posting	VERIFIED SAFE (Emerald)	Authentic corporate domain, verified skills, structured interview, and fair stipend.
Unrealistic 80k Monthly Pay Bait	UNREALISTIC PAY (Amber)	Rs. 80,000/mo for 2hrs/day mobile work; zero prerequisites or resume screening.
Urgent WhatsApp Contact Flyer	URGENCY PRESSURE (Orange)	Artificial pressure ('Only 3 spots left tonight') diverting applicants exclusively to WhatsApp.
Informal Early-Stage Startup Post	INFORMAL STARTUP (Purple)	Small team with personal blog; moderate caution needed without upfront fee trap.

6. Viva Defense Quick Reference

Q1: Why was Linear SVM chosen over Deep Learning (BERT/LLMs)?

Linear SVM on TF-IDF sparse features achieves 98.41% accuracy and 0.8190 F1-score with sub-5ms inference latency on CPU, with zero GPU cost, API dependencies, or token bills.

Q2: How does the system handle class imbalance in recruitment datasets?

Only ~4.8% of the EMSCAD corpus is fraudulent. We applied balanced class weighting and calibrated probabilities using Sigmoid cross-validation (CalibratedClassifierCV) to prevent majority-class bias.

7. Project Submission & Deliverables Checklist

- ✓ **GitHub Repository:** <https://github.com/vaibhavmurty4518/fake-internship-detector.git> (Up to date on main)
- ✓ **Submission Zip Archive:** C:\Users\Vaibhav Murty\Downloads\fake-internship-detector-review3-final.zip
- ✓ **Automated Test Suite:** 48 / 48 Tests Passed (100% Pass Rate via Pytest)
- ✓ **Zero Code-Leak Guarantee:** Complete custom HTML rendering stream verified across all 8 tabs.